Geometric Formulation of Spacetime via Central Projection: Overview of 10 Papers

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Paper List

#	Title	DOI
[1]	Geometric Formulation of 4-Dimensional Space	10.5281/zenodo.19427780
[2]	Geometric Symmetries of Central Projection	10.5281/zenodo.19434932
[3]	Relativity of Observation in Multiple Subjective Spaces	10.5281/zenodo.19435162
[4]	On the Limits of the Curvature Radius	10.5281/zenodo.19526549
[5]	On Dimension Addition to Background/Subjective Spaces	10.5281/zenodo.19526913
[6]	Subjective Spaces from Zero to Four Dimensions	10.5281/zenodo.19533292
[7]	On the Continuity of Geodesics in Subjective Spaces	10.5281/zenodo.19533299
[8]	Circumscribed Hypersphere of a Unit 4D Hyperrectangle	10.5281/zenodo.19533313
[9]	Schwarzschild–de Sitter Exact Solution	10.5281/zenodo.19538098
[10]	Dimensional Interpretation at the $R \to 0$ Limit	10.5281/zenodo.19538106

Purpose of This Research

From purely geometric properties of central projection from an n-dimensional tangent hyperplane to a hypersphere $S^n(R)$, we construct a unified framework in which multiple independent axes that generate acceleration can coexist without contradiction. We refrain from physical identifications and confine ourselves to geometric propositions and their proofs.

Proven Geometric Propositions

Paper [1]: Basic Structure of Central Projection

1. Isomorphism with de Sitter Spacetime The induced metric of the Lorentzian central projection coincides numerically with the metric in the Beltrami coordinate system

of de Sitter spacetime, establishing complete equivalence as intrinsic geometry. (§1.1, Appendix C, Proposition C.1)

2. Regularity of Forward and Inverse Projection Since $l = \sqrt{R^2 + r^2} \geq R > 0$, the central projection Φ is C^∞-regular on its entire domain. The inverse $x^\mu = RY^\mu/Y^{n+1}$ is regular on H^+. The mapping is bijective. (Appendix B, Proposition B.1)

3. Induced Metric and Einstein Tensor The pullback metric $g_{\mu\nu} = (R^2/l^2)(\delta_{\mu\nu} - x_\mu x_\nu/l^2)$ is rigorously derived, and the n-dimensional Einstein tensor satisfies $G_{\mu\nu} + \Lambda_n g_{\mu\nu} = 0$ with $\Lambda_n = (n-1)(n-2)/(2R^2)$. (§3, §6, Proposition 6.1)

4. Flat Limit As $R \to \infty$: $g_{\mu\nu} \to \delta_{\mu\nu}$, $\Lambda \to 0$. The central projection degenerates to the identity map, recovering Minkowski spacetime. (Appendix A)

Paper [2]: Four Geometric Symmetries

5. Symmetry I (Discrete Stability) For any coordinate value — zero, positive integer, or negative integer — $l_A > 0$ holds and the central projection is globally regular. (§3, Theorem 3.1)

6. Symmetry II (Axis Equivalence) Regardless of which axis in $\mathbb{R}^{n+1}$ is chosen as the projection center, the metric, curvature tensor, and Einstein tensor have identical mathematical structure. (§4, Theorem 4.1)

7. Symmetry III (Geodesic Deviation and Indistinguishability) The geodesic deviation $|\xi|^2 = g_{\mu\nu}\xi^\mu\xi^\nu$ is a scalar invariant of dimension $[L^2]$ and the sole means by which an internal observer recognizes curvature. However, neither the sign of coordinates nor the direction of curvature can be distinguished. (§5, Theorem 5.2)

8. Symmetry IV (Transformability of Subjective Coordinate Systems) Subjective coordinate systems centered on different axes are mutually transformable via $T_{A\to B} = \Phi_B^{-1} \circ \Phi_A$, a C^∞ diffeomorphism. The only parameters are the curvature radius R and coordinate values. (§6, Theorem 6.1, Proposition 6.5)

9. Curvature Radius as Scale Factor The Jacobian is $\partial x'^A/\partial x^B = -R^2/(x^B)^2$, with a dimensionless first-order scale factor. The curvature radius R serves as the sole transformation parameter. (§6.4, Proposition 6.2)

Paper [3]: Multiple Subjective Spaces and Relativity of Observation

10. Simultaneous Existence of Multiple Subjective Spaces From $\mathbb{R}^{n+1}$, $n+1$ subjective spaces $\{H_A^+\}$ can be simultaneously constructed. Each possesses structurally identical geometric properties by Symmetry II. (§2, Proposition 2.1)

11. Limits of Information Obtained by Internal Observers An internal observer obtains only quantities from the metric of their own subjective space and cannot access the projection center axis value, absolute position, or the existence of other subjective spaces. (§3, Proposition 3.1)

12. Exchange of Axis Roles and Relativity of Observation What is the "projection center axis" for observer A appears as one of the subjective coordinates for observer B, and vice versa. No geometric basis exists for determining which standpoint is "correct." (§4, Proposition 4.1; §5, Theorem 5.1)

Paper [4]: Limits of the Curvature Radius

13. Regularity of the $R \to \infty$ Limit As $R \to \infty$, the induced metric asymptotes to flat Euclidean space and $\Lambda = 3/R^2 \to 0$. All terms of the Einstein equation exhibit well-defined asymptotics. (§2.1)

14. Regularity and Structural Tension at $R \to 0$ As $R \to 0$, $\Lambda = 3/R^2$ diverges, but all terms remain well-defined for any $R > 0$. The upper bound of geodesic deviation approaches zero while local curvature diverges at order $1/R^2$, giving rise to structural tension. (§3.1, §3.2)

15. Existence of Nested Structure For any point p in any subjective space Π_R, a new subjective space $\Pi_{R'}$ with p as the elevation point exists for any $R' > 0$. (§4.2, Proposition 4.1)

Paper [5]: Dimension Addition

16. Two Categories of Dimension Addition Type I (under existing curvature radius R) and Type II (introducing a new independent curvature radius R'). (§2.2)

17. Dimension Addition to Finite-Curvature Subjective Spaces Type I is freely executable. For Type II, unless the new axis and governed dimensions are added simultaneously, the subjective space degenerates to zero dimensions. (§4.2, Observation 4.1; §4.3, Observations 4.2–4.3)

Paper [6]: Subjective Spaces from Zero to Four Dimensions

18. Characteristics at Each Dimension 0D: No internal structure. 1D: No direction concept. 2D: Area, angle, geodesic deviation first definable; time axis undetermined. 3D: Volume definable. 4D: Hypervolume definable; number of time axes not uniquely determined. (§2–§6)

Paper [7]: Continuity of Geodesics

19. Independence of Parent-Child Geodesics A geodesic γ on Π_R and a geodesic γ' on child $\Pi_{R'}$ (elevated from a point on γ) are independently defined geometric objects sharing the same ambient space. (§4.3, direct consequence of Theorem 6.1)

Paper [9]: Schwarzschild–de Sitter Exact Solution

20. Unique Construction of the Exact Solution The spherically symmetric static exact solution of the vacuum Einstein equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ ($\Lambda = 3/R^2$) from Paper [1] is uniquely determined as the Schwarzschild–de Sitter metric $f(r) = 1 - 2M/r - r^2/R^2$ by Birkhoff's theorem. No additional assumptions required. (§3, Eq. 3.4)

21. Disappearance of Horizon Structure at $R \to 0$ As $R \to 0$, the cosmological horizon $r_{\mathrm{C}} \to 0$, merging with the black hole horizon and eliminating the observer region. This is the exact-solution-level counterpart of Paper [4] §3.2. (§4.2–§4.3)

22. Recovery of Horizon Structure in Child Subjective Space By Proposition 4.1, a child subjective space $\Pi_{R'}$ with independent R' can be constructed, and the inter-horizon observer region recovers when R' is finite. (§5.2–§5.3)

Paper [10]: Dimensional Interpretation of the $R \to 0$ Limit

23. Correspondence between $R \to 0$ and Zero-Dimensional Subjective Space The $R \to 0$ limit corresponds geometrically to Paper [6]'s zero-dimensional subjective space: disappearance of geodesic deviation, internal structure, and curvature recognition. (§3, Observation 3.1)

24. Description as Dimensional Transition Stage 1 (Dimensional collapse): parent's dimensional structure is lost, approaching zero dimensions. Stage 2 (Dimensional reconstruction): child subjective space is elevated via Proposition 4.1 and dimensional structure unfolds. The geodesic "terminus" is a transition point, not annihilation. (§5, Observation 5.1)

Paper [8]: Singularity of Four Dimensions

25. Integer Property of the Circumscribed Hypersphere Diameter The diameter $2\sqrt{n}$ of the circumscribed hypersphere of a unit n-dimensional hypercube is an integer iff n is a perfect square. Excluding trivial $n = 1$, $n = 4$ is the smallest non-trivial dimension satisfying this. (§3)

26. Settling at Four Dimensions The dimension simultaneously satisfying symmetry preservation, integer property, and minimization conditions is $n = 4$. (§12)

27. Origin of the Time Axis An odd number of axis inversions is required for negative volume. The most economical single-axis inversion generates $(-1)^1 = -1$. This "1-to-3 asymmetry" gives rise to one time axis and three spatial axes in $(-, +, +, +)$. (§13)

What This Research Does Not Claim

This research is confined to purely geometric propositions and their proofs. The following claims are not made:

- That internal observers of subjective spaces physically exist
- That "relativity of observation" corresponds to any specific theory of relativity in physics
- That a particular subjective space corresponds to our universe
- That physical laws are derivable from the structure of subjective spaces

These physical interpretations require additional assumptions beyond the geometric propositions of this research (Paper [3], §7).

References

[1] Kihara, N. (2026). *Geometric Formulation of 4-Dimensional Spacetime via Gnomonic Projection*. DOI: 10.5281/zenodo.19427780.

[2] Kihara, N. (2026). *Geometric Symmetries of Central Projection*. DOI: 10.5281/zenodo.19434932.

[3] Kihara, N. (2026). *Relativity of Observation in Multiple Subjective Spaces*. DOI: 10.5281/zenodo.19435162.

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